WORKSHOP

GLOBAL ENVIRONMENTAL ENERGY TECHNOLOGY

OFFICIAL GEET SMALL ENGINE PLANS

The following plans are designed for use on small engines, such as those used on lawnmowers, small generators, pumps, etc. The diagram drawings represent working designs of basic individual parts and components needed for a manually operated retrofit application. Changes or variations of design may be necessary to fit your specific needs (example: flange modifications, parts placement, space limitations etc.). As long as the principles of the GEET technology are adhered to and all system components are incorporated as specified, your retrofit application should function properly.

WARNING!! - This information is classified as EXPERIMENTAL!! We cannot guarantee results. Please take time to educate yourself sufficiently before proceeding. The following is for information purposes only. We do not control the materials used in construction, the methods of construction, nor the applications of construction or use of the finished product. Therefore, WE ARE NOT RESPONSIBLE FOR DANGEROUS OR UNDESIRABLE RESULTS, and we do not take responsibility for accidents due to negligence or ignorance!!

WARNING!! - Gasoline is EXTREMELY FLAMMABLE and caution should always be used when working around it. Do not smoke around it and keep it away from open flames. Whenever possible use goggles, gloves and/or other protective gear.

WARNING!! - DO NOT COPY, FAX, REDRAW OR USE ANY OTHER FORM OF REPRODUCTION! PLANS ARE FOR PERSONAL USE ONLY. ANY VIOLATION WILL BE PROSECUTED UNDER PENALTY OF LAW!!

WHAT IS A GEET FUEL PROCESSOR?

The GEET Fuel Processor (GFP) is a fuel refinery, plasma field generator, and fuel delivery system all rolled into one. The GFP is capable of dramatically increasing the efficiency and decreasing the pollution of an internal combustion engine. The GFP is also capable of using diverse fuels that the engine would normally not tolerate. For many applications, the GEET retrofit can pay for itself in a short period of time. To better understand how the GFP works, let's look at the smaller sub-systems, of which there are three.

1)Liquid-to-Vapor Phase Changer - The GFP reactor utilizes a vaporized fuel, not a liquid fuel. Therefore we must incorporate a means to change the liquid fuel, by means of a phase change, into a vapor. The drier the vapor the better. Realistically, a bubbler or even a modified carburetor can, at best, deliver an aerosol, which is small droplets of liquid fuel mixed with air. The heavier the fuels the more difficult it is to vaporize them. Between the two, the bubbler has delivered the better results, in the use of heavier or diverse fuels. The bubbler also seems to be more efficient in the vaporization of fuel, but the modified carburetor is easier to work with, and has its own float system built in. For on-going operation, the bubbler will need to have a float system setup to regulate the level of fuel in it.

2) Reaction Chamber - This is a pipe within the exhaust system which utilizes a heat transfer or plasma reaction to break the fuel down into a simpler, cleaner form before it is fed into the engine. By running fuel vapor up through the center of the exhaust in the opposite direction of the exhaust flow we are simulating a natural phenomena of hot and cold air masses colliding. This collision within a vacuum causes the desired "reaction" we are looking for. The following three effects must be present in the reaction chamber in order to create the reaction (assuming the hardware is correct)"

a) Heat
b) Vacuum on the inner chamber
c) Cross flow (hot in one direction, cold in the other).

Upon initial start-up you will have vacuum and cross flow, but not heat. Therefore, you must start the engine on a suitable fuel such as gasoline until the engine and reaction chamber are at normal operating temperatures. This can be accomplished by either putting sufficient gasoline in the fuel you desire to run, or using a separate tank for start up purposes. A valve can be placed in the fuel line to switch from start up fuel to what ever fuel you desire to run on. Our advice is to try and KEEP IT SIMPLE!

3) Air Management Valve (AMV) - The new "GEET Fuel" coming from the reaction chamber must be properly mixed with additional incoming air, then metered into the engine as needed. This is the job of the air management valve. This can be as simple as using two ball valves or as complicated as machining your own design.

Within these plans, we have diagramed a manual operated air management valve. Because of the wide variety of engines, parts, etc, we are not able, at this time, to provide a diagram of an air management valve which is controlled by the engine, and not manually.

Understand that you have the freedom to mix and match styles and designs to best suit your application, space requirements, and abilities. For security and protection reasons, we are not including the specific clearances for each application. You will need to call GEET Management at (801) 281-4577 to get these specifications. When built properly, the GFP should work as promised and in most instances may exceed your expectations.

THE BUBBLER

One of the two methods of phase changing the liquid fuel into a vapor is a bubbler. A diagram of a simple bubbler design is included in the plans. A manual valve is used to adjust the amount of air that is fed into the bubbler. For proper operation, the valve should be more closed than open to allow a vacuum to be maintained within the bubbler. As that air bubbles up through the liquid fuel, it creates vapors in the top portion of the bubbler which are pulled into the reaction chamber by the vacuum of the engine. A diffuser/bubbler plate and/or metal "scrub pads" placed in the bubbler will help break up the larger bubbles into smaller ones to produce better vapors, enrichening the fuel mixture to the reaction chamber. Various other types of bubblers can be used also. The liquid level within the bubbler should never be high enough to allow the liquids to be sucked into the reaction chamber. Only the vapors are allowed in the reaction chamber for the process to work properly. The heavier fuels will require the bubbler to be heated to enhance their full vaporization. This can be accomplished by either surrounding the bubbler with hot exhaust from the engine, and/or directing some of the exhaust into the bubbler instead of fresh air. The bubbler can be constructed of metal or some plastics such as PVC. When using plastics, make sure the glue used will not be dissolved by the fuels you will be putting into the bubbler. We have found that J B Weld epoxy is good to use. If you choose to use exhaust gases to surround and heat the bubbler, then plastics should be avoided due to the hot temperature of the exhaust. If you choose to use exhaust gases to bubble the fuel with, there are several factors to consider. First, when tapping into the exhaust system, come out in a "Y " fashion. If you "T" into the exhaust, there could be a siphoning effect that might suck the fuel into the exhaust. By coming out of the exhaust at an angle ("Y") under a slight pressure, the siphoning problem is eliminated.

Secondly, the line carrying the exhaust to the bubbler must at some point rise above the top of the bubbler to eliminate drainage back into the line upon shutdown.

Thirdly, you must use the " manual valve" between the exhaust source and the bubbler inlet to control the amount of exhaust entering the bubbler to maintain a vacuum. Having the valve too far open could cause the vacuum to be lost and a "pressure" to be formed in the bubbler. We have noted in our testing that efficiencies can be cut to 1/3 by pressurizing the bubbler (as opposed to having it under a vacuum). The valve must also be closed when starting or stopping the unit to prevent fuel from being forced into the exhaust by back pressure from the engine.

Finally, all of the components exposed to the heat of the exhaust must be capable of handling this heat. Most ball valves use a Teflon seal inside that will melt with the heat of the exhaust. Furthermore, if using a PVC bubbler, the exhaust would melt the plastic at the point of inlet. We have used a metal screw-in lid in both the "top" and "bottom" applications of admitting a bubbling effect into the bubbler. The metal lid will dissipate the heat into the fuel before it comes in contact with the PVC. When using the "bottom" method, for bubbling, make sure the inlet line or tube at some point rises higher than the bubbler to prevent the liquid fuel from running back out upon shutdown of the engine. The "valve" should be placed at that "higher" spot also.

In summary, the bubbler can be square, round, heated, not heated, tall or short (make sure it is tall enough to keep the liquids from being sucked into the reaction line!). Whatever the specific needs of your application are will determine if you should use a bubbler and how that bubbler should be built. For constant running applications, a means of controlling the level of fuel (float chamber, external electric float unit or something that your design yourself) should be used. The large bubbles coming into the bubbler need to be broken down into smaller bubbles by the use of a bubbler/diffuser plate and/or pot scrubbers. Be sure to use materials able to handle the fuels and temperatures that they will be exposed to. Either fresh ambient air or exhaust gasses can be bubbled up through the bubbler. The ambient air is easier to work with, but the exhaust gasses may offer better results.

MODIFIED CARBURETOR

For certain applications a bubbler may be impractical. Space limitations or other restrictions may require the use of a modified small engine carburetor. We have successfully used 8- and 10- HP carburetors of horizontal shaft Tecumseh engines. The drawing shows that we restrict the venturi in the carburetor to pass more air over the pick up tube.

The stock carburetor is designed to deliver a 12:1 air fuel ratio (AFR). For the GEET we need a 2 or 3:1 AFR. This is achieved by restricting the venturi diameter and forcing all of the air past the fuel feed tube. This can be accomplished by either drilling a small hole on the choke side of the venturi and mounting a baffle to direct all of the air over the discharge feed tube (#5), or by using JB Weld (epoxy) to "secure" a washer into the venturi, restricting the inlet (leaving the choke on full may also help channel the air). Combinations of all of these tricks can make the mixture richer to deliver the necessary fuel charge for a strong and powerful reaction.

The mixture adjustment screw (#6 on diagram) can also be used to regulate the amount of fuel to be delivered to the reactor, helping control engine power requirements. NOTE: If you adjust the Air/Fuel mixture going to the reaction chamber too much on the "rich" side, economy will decrease and pollution levels will increase.

THE REACTION CHAMBER

The heart of the GEET Fuel Processor is its reaction chamber. In a sense, it is the single most important part of the entire system. Ironically, it is also usually the easiest part to build and tune! Most of the time if a system isn't operating properly it isn't the fault of the reactor (unless there are leaks). Improper fuel mixtures, vacuum leaks, too much liquid being fed into the reactor, or other things will cause problems before the reactor will.

The reaction chamber and rod need to be tuned to the size of the engine, the fuel being used, the RPM range of intended use, and whether you are using a single or dual reactor system.

Generally the exhaust pipe which surrounds the reactor chamber (pipe) should be sized large enough so as not to restrict the flow of exhaust from the engine. The diagram illustrates that the exhaust can be bent with the inner pipe entering and exiting straight, or the exhaust can go straight with the inner pipe entering and exiting through elbows.

Steel exhaust pipe is recommended for the outer part of the reactor. Stainless steel should be avoided in all parts of the reactor. All of the parts should be of a ferrous metal; or in other words, should be easily attracted to a magnet. This means no copper, brass, aluminum, or stainless.

Low grade steel dowel is what we use for the rod. The length and diameter of both inner pipe and reaction rod will only be given over the phone for the application you are building. This is done for our protection.

Remember that your exhaust gasses will be exiting the engine and traveling in one direction, while the fuel will be traveling up the inner pipe in the other direction. Hot in one direction, cold in the opposite direction.

The GFP reaction chamber has shown to generate its own electromagnetic field (EMF). If applying your GFP to a generator, route the reactor away from the generator. The EMF from the generator may interfere with the performance of your GFP.

On V-configurated engines you can run a single reaction chamber, or run dual. If you have the room, the twin set-up may be the preferred method. If you don't have enough room, mount the reaction chamber down stream of the "Y" pipe. This enables you to utilize all of the engine's waste exhaust heat.

The reaction rod has three equally spaced tabs welded onto it at both ends, within 1/2" of the end. Either "bullet" shape the end of the rod, or grind grooves between the tabs to create better aerodynamics for the incoming fuel.

AIR MANAGEMENT VALVE

Of all the aspects of building the GFP, the air management valve (AMV) is the part that may give you the most trouble when you try to adapt or build one to work "automatically" instead of manually. The function of the AMV is two fold; first to control the amount of air and fuel, and secondly to control the engine speed and power output.

It is recommended that you use manual controls at first to further understand the workings of the GFP. Once you are familiar with how the engine operates with the GFP system it will be easier for you to understand what changes will be necessary for automatic controls.

Use a ball valve for the air inlet to the bubbler, and a ball valve for the air bleed (see illustration). As you open the fuel valve, open the air valve too. To lower engine speed, slowly close both valves simultaneously.

For a stationary engine, constant load application, the ball valves can be the easiest way to go. Use one for the bubbler inlet (fuel control) and one for the ambient air inlet.

The air valve is like the throttle butterfly on any conventional fuel delivery system. We want to create a 2:1 or 3:1 air fuel mixture at our bubbler or modified carburetor. We will admit the remaining air at the air valve.

Drawing labeled "Small Engine Air Management Valve" was used on our "Dealer Demo Engines". Using this also required a bubbler valve. This is a manual (not automated) operation piece. Not recommended for other than learning purposes.

PUTTING IT ALL TOGETHER

Big, small, or someplace in between, all of the necessary elements must come together into a usable package. To review, there are three sub-assemblies:

1) The phase change mechanism, which is a bubbler, or a modified carburetor.

2) The reaction chamber, where the "magic" happens.

3) The air management valve which controls the fuel/air ratio and engine speed.

Certain things cannot be related on paper. You will learn more by running the engine for the first time then could be taught on a hundred pages of print. Remember, don't hesitate to call with your questions. We would rather spend 15 minutes on the phone and have you get it right than being frustrated and unhappy with our products.

SUMMARY

Believe it or not, the basic GEET system isn't too overly critical of sizes and dimensions. It is extremely critical of vacuum leaks. If things aren't working well for you, check for any vacuum leaks.

Start out on something small. Start out with manual controls and learn how the GEET works. Don't hesitate to call at (801) 281-4577 or fax your questions at (801) 281-4578 or E-Mail to <info@GEET.com>.

GOOD LUCK

Storage tank is kept as an appropriate pressure by the Compressor and sensors

The Carbuerator is designed to mix a flow of GEET gas from the presure tank, and air.

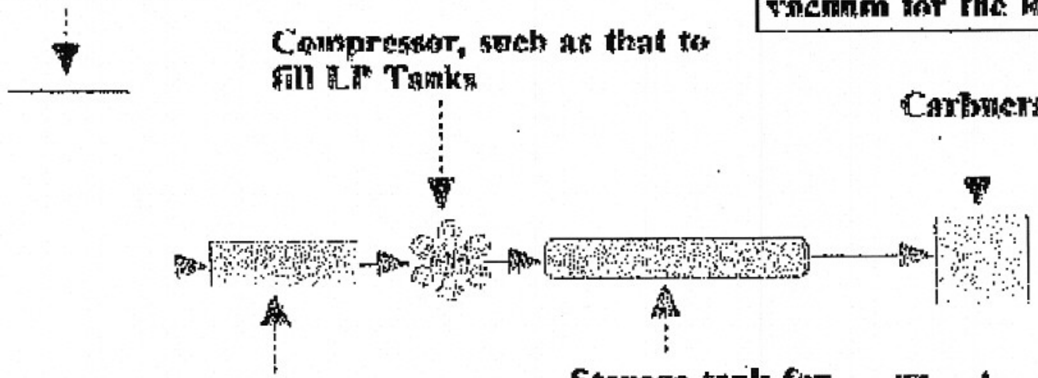
The GEET Reactor is designed to supply enough Gas flow, as to exceed the maximum used by the engine.

The Compressor creates the vacuum for the Reactor

Alternate Fuel

Compressor, such as that to fill LP Tanks

Carbuerator



GEET Reactor

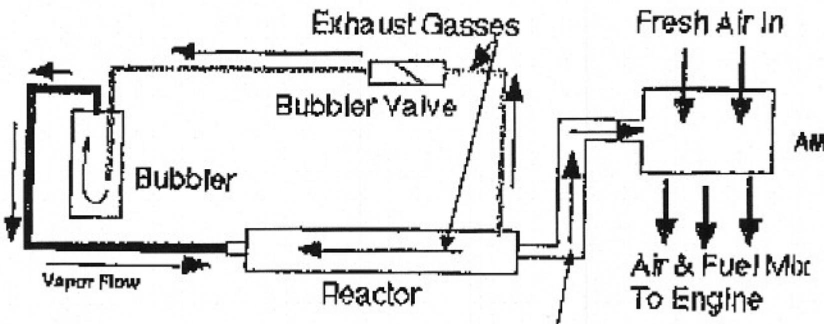
Storage tank for GEET gas-
This could be a large pipe with caps drilled to connect to Reactor and Carbuerator

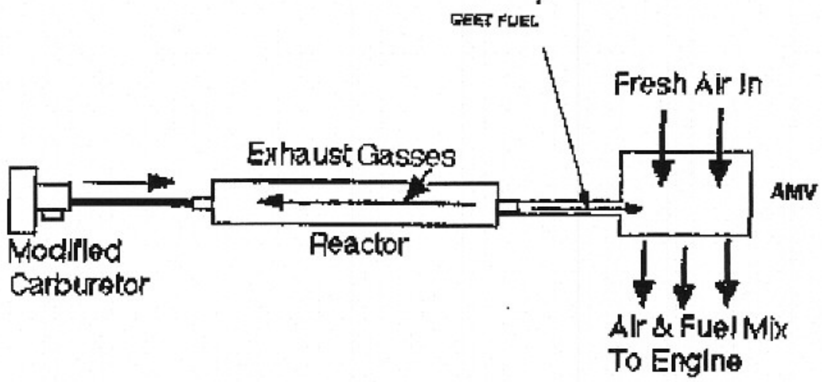
The size of the tank and the pressure should be researched to combine the best dimmension and pressure for stable operation

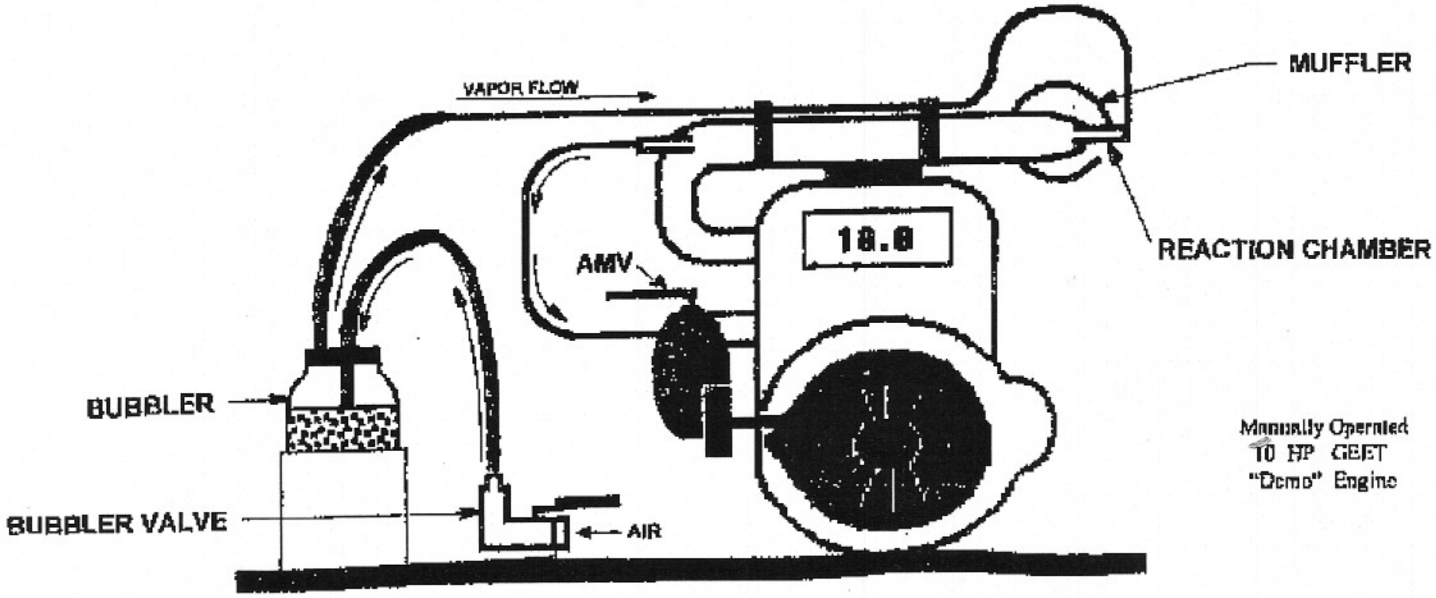
A Design by
L. Gerald Buttle
July 22, 2001
15:02

L Gerald Buttle

General Overview



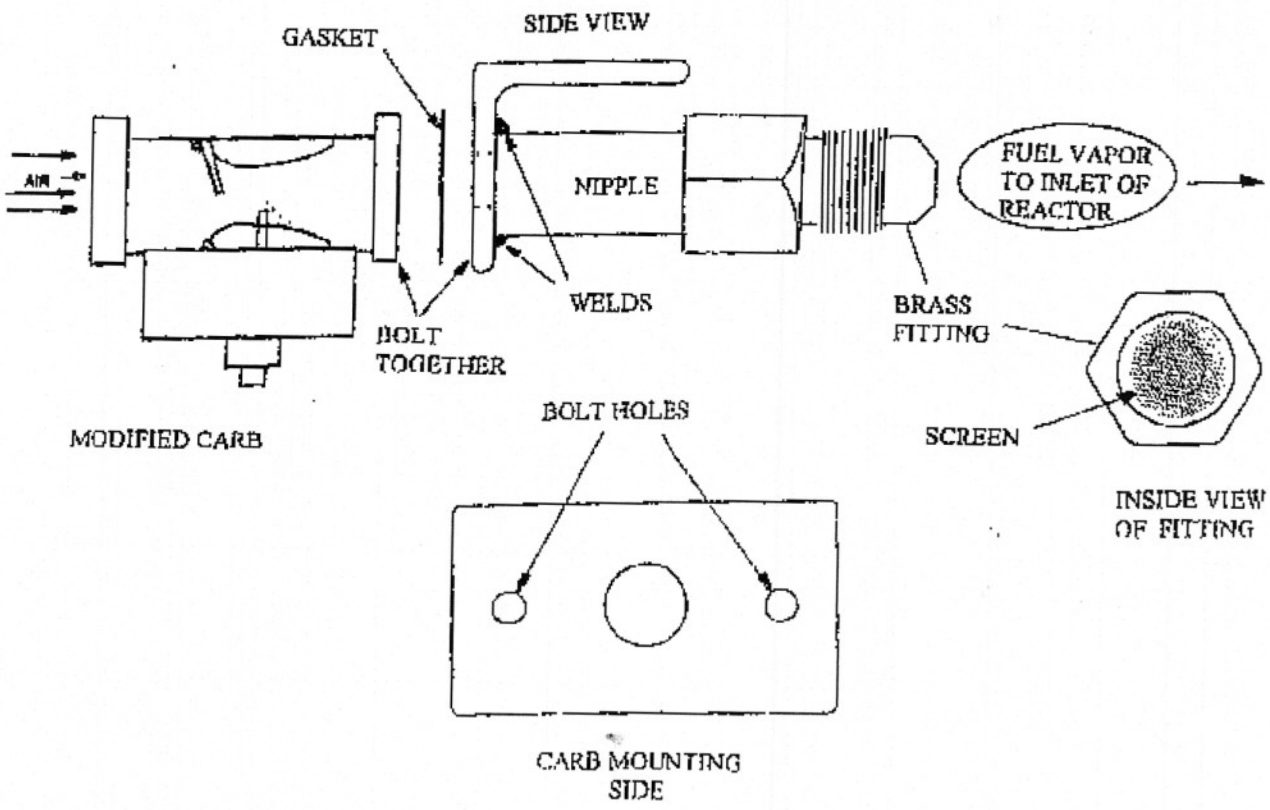




Manually Operated
10 HP GEET
"Demo" Engine

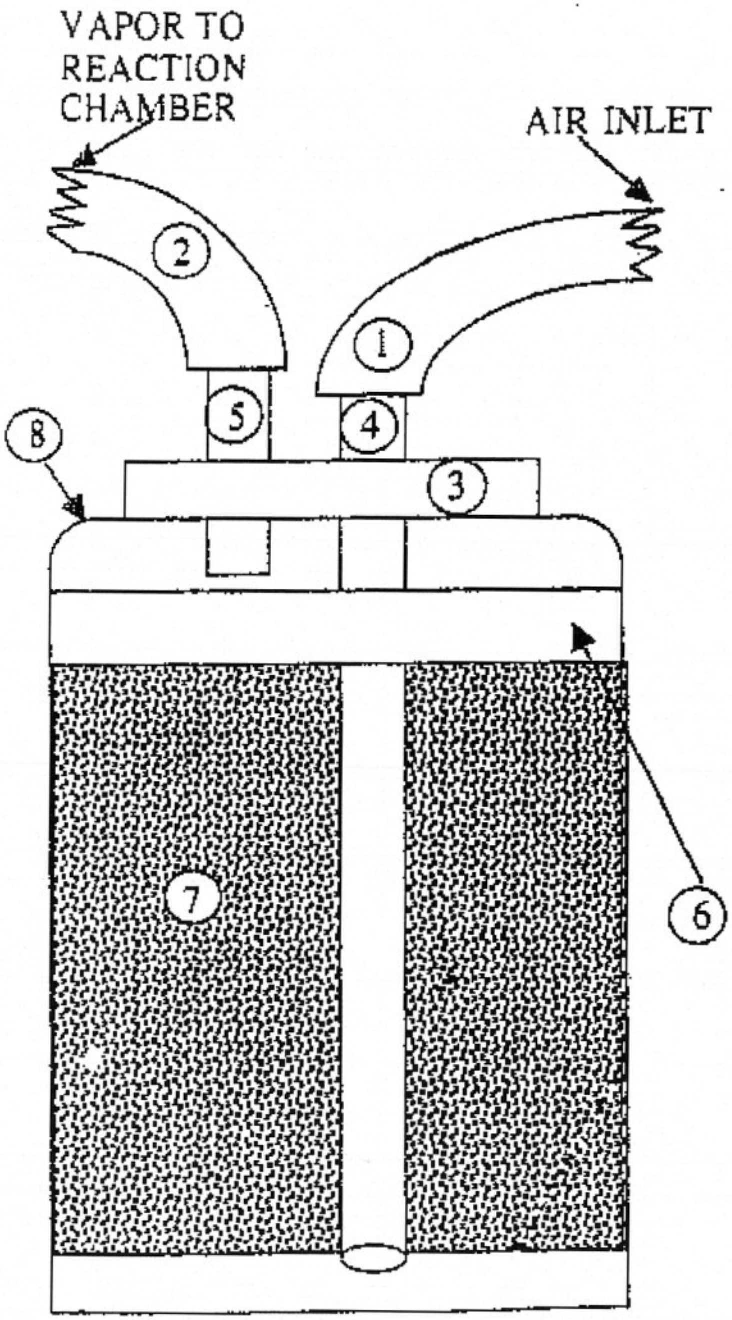
MODIFIED CARBURETOR MOUNTING FLANGE

If you choose to use a modified carburetor instead of a bubbler you will need to decide where to place it and how to connect the necessary lines for fuel in and Air/Fuel mixture out. You will be using the float bowl system on it to constantly control the fuel flow from the tank, so remember that the fuel tank <u>must</u> be higher than the carburetor. A simple mounting bracket like the diagram below can be built from the appropriate size "angle Iron". Flat steel may also be used, depending on how you decide to mount it. Use bolts and locking nuts to connect the carb (also use a gasket) to the bracket. On the other side of the bracket weld a 1/2" x 1 ½ - 2" NPT steel nipple for fuel/air mixing and connecting to your 1/2" tubing which goes to the reaction chamber. Before you screw the brass fitting on the end of the nipple, place a small piece of fine brass screen inside the fitting so it covers the opening (make sure it stays secure). This is a safety precaution we use in case the engine should backfire. Once you have secured the completed unit to your engine, connect the tubing from the carb to the reaction chamber. **Note:** The more angles and restrictions you place between the fuel/vapor going to the reaction chamber and then to the air management valve, the more difficult it will be to get the fuel to the engine. The easier the flow, the better.

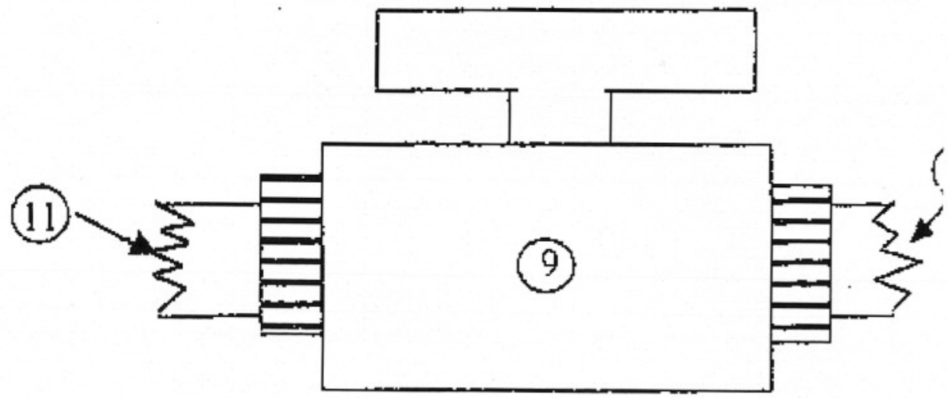


NOT) SCALE

MASON JAR BUBBLER USED ON "DEMO" ENGINE



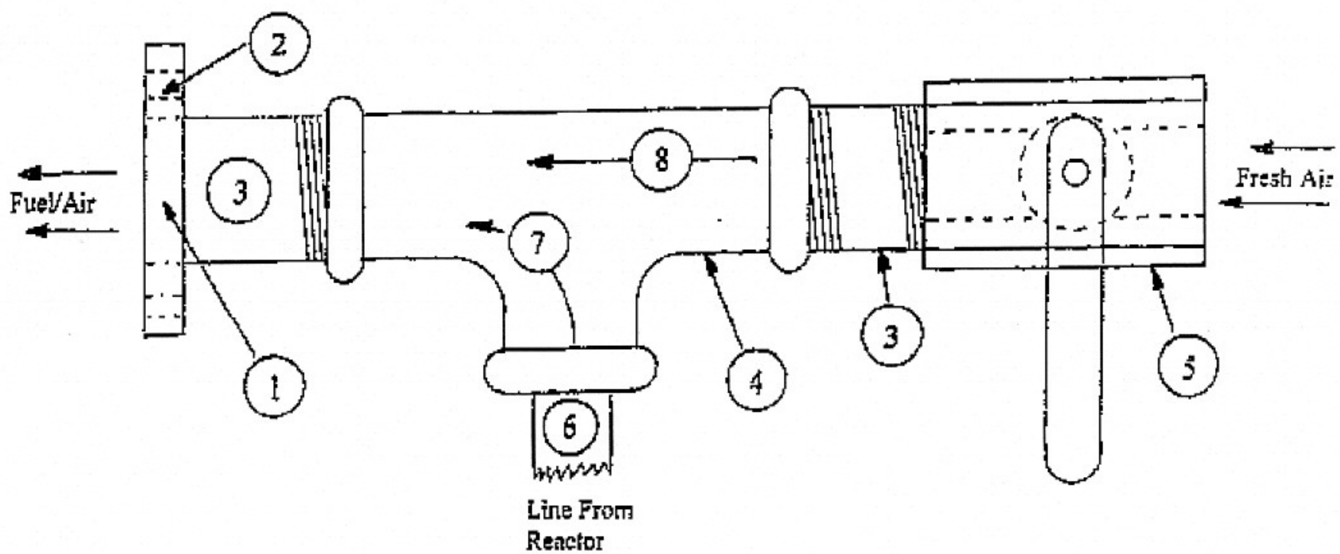
1) 1/2" FUEL LINE TO VALVE #9
2) 1/2" FUEL LINE TO REACTION CHAMBER
3) PLASTIC CORK TO FIT MASON RING
4) 1/2" COPPER TUBING
5) 1/2" COPPER TUBING(DOES NOT GO BELOW #6)
6) AIR SPACE ABOVE #7 MATERIAL
7) POT SCRUB PADS OR STEEL WOOL
8) MASON JAR
9) BALLCOCK VALVE TO CONTROL AIR INLET
10) LINE TO AIR FILTER
11) LINE TO AIR INLET OF BUBBLER POT



AIR MANAGEMENT VALVE ALTERNATIVE

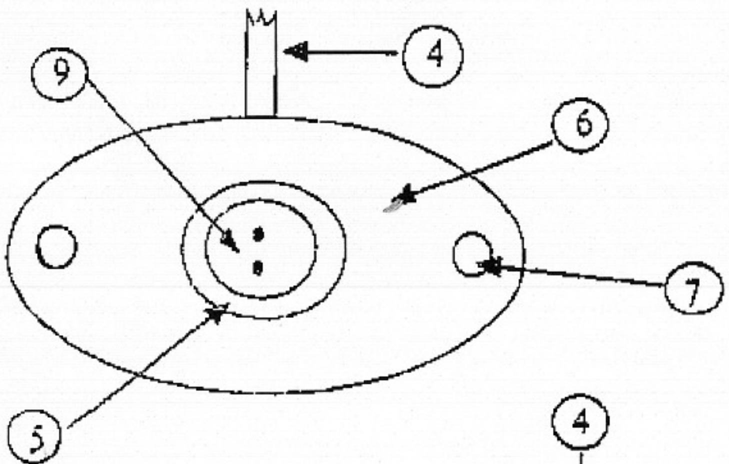
AS AN ALTERNATIVE TO "MACHINING" AN "AMV" AS DESCRIBED ON THE OTHER SIDE, YOU CAN CONSTRUCT A SIMPLE UNIT FROM PLUMBING PARTS FOUND AT YOUR HARDWARE STORE. THIS MAY BE A'MORE DESIRABLE METHOD TO START WITH.

1) FLANGE TO BOLT TO ENGINE INTAKE
2) BOLT HOLES IN FLANGE
3) 1/2" or 3/4" NIPPLES (depending on engine HP)
4) STEEL OR BRASS "T"
5) 1/2" or 3/4" BALL VALVE
6) BRASS FITTING FOR 1/2" TUBING
7) FUEL FROM REACTOR
8) FRESH AIR

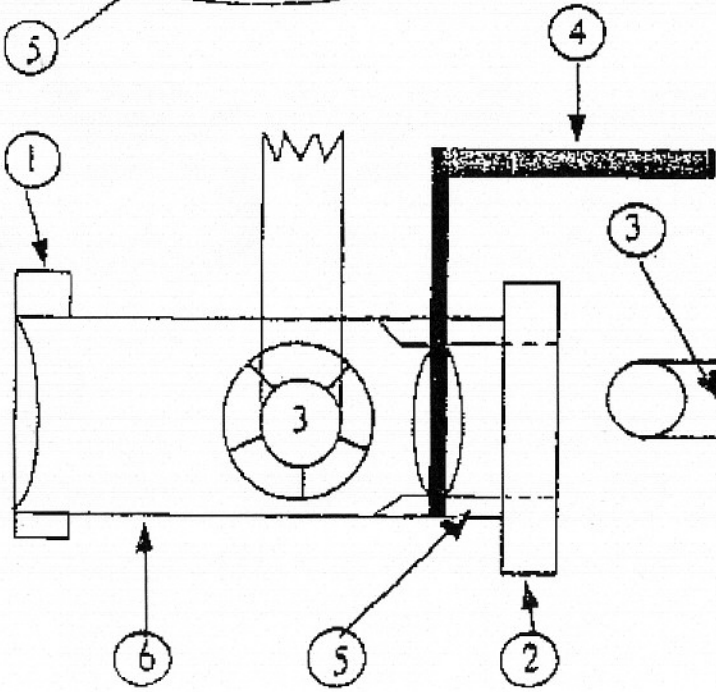


SMALL ENGINE AIR MANAGEMENT VALVE
(AMV)

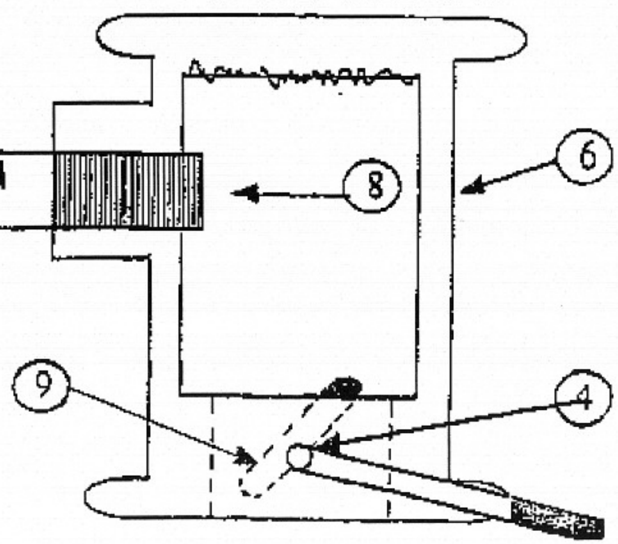
END VIEW OF AIR FILTER SIDE



1) MOUNTING FLANGE TO INTAKE MANIFOLD
2) MOUNTING FLANGE TO AIR FILTER
3) COMPRESSION FITTING FOR "GEET FUEL" FROM REACTION CHAMBER
4) THROTTLE HANDLE TO BUTTERFLY VALVE
5) MACHINED VENTURI
6) AMV MACHINED FROM ALUMINUM OR STEEL
7) MOUNTING BOLT HOLES
8) FUEL INLET HOLE DRILLED AND THREADED
9) BUTTERFLY VALVE



TOP VIEW / CUT AWAY



NOT TO SCALE